

7. The line  $l_1$  has equation

$$\mathbf{r} = \mathbf{i} - 2\mathbf{j} + 3\mathbf{k} + \lambda(2\mathbf{i} + \mathbf{j} - 4\mathbf{k})$$

and the line  $l_2$  has equation

$$\mathbf{r} = 5\mathbf{i} + p\mathbf{j} - 7\mathbf{k} + \mu(6\mathbf{i} + \mathbf{j} + 8\mathbf{k})$$

where  $\lambda$  and  $\mu$  are scalar parameters and  $p$  is a constant.

The plane  $\Pi$  contains  $l_1$  and  $l_2$

- (a) Show that the vector  $3\mathbf{i} - 10\mathbf{j} - \mathbf{k}$  is perpendicular to  $\Pi$  (2)
- (b) Hence determine a Cartesian equation of  $\Pi$  (2)
- (c) Hence determine the value of  $p$  (2)

Given that

- the lines  $l_1$  and  $l_2$  intersect at the point  $A$
  - the point  $B$  has coordinates  $(12, -11, 6)$
- (d) determine, to the nearest degree, the acute angle between  $AB$  and  $\Pi$  (4)



7. The line  $l_1$  has equation

$$\frac{x-1}{2} = \frac{y+1}{-1} = \frac{z-4}{3}$$

The line  $l_2$  has equation

$$\mathbf{r} = \mathbf{i} + 3\mathbf{k} + t(\mathbf{i} - \mathbf{j} + 2\mathbf{k})$$

where  $t$  is a scalar parameter.

(a) Show that  $l_1$  and  $l_2$  lie in the same plane.

(3)

(b) Write down a vector equation for the plane containing  $l_1$  and  $l_2$

(1)

(c) Find, to the nearest degree, the acute angle between  $l_1$  and  $l_2$

(3)

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4. The plane  $\Pi_1$  has equation

$$\mathbf{r} = 2\mathbf{i} + 4\mathbf{j} - \mathbf{k} + \lambda(\mathbf{i} + 2\mathbf{j} - 3\mathbf{k}) + \mu(-\mathbf{i} + 2\mathbf{j} + \mathbf{k})$$

where  $\lambda$  and  $\mu$  are scalar parameters.

- (a) Find a Cartesian equation for  $\Pi_1$  (4)

The line  $l$  has equation

$$\frac{x-1}{5} = \frac{y-3}{-3} = \frac{z+2}{4}$$

- (b) Find the coordinates of the point of intersection of  $l$  with  $\Pi_1$  (3)

The plane  $\Pi_2$  has equation

$$\mathbf{r} \cdot (2\mathbf{i} - \mathbf{j} + 3\mathbf{k}) = 5$$

- (c) Find, to the nearest degree, the acute angle between  $\Pi_1$  and  $\Pi_2$  (2)

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7. The plane  $\Pi$  has equation

$$\mathbf{r} = \begin{pmatrix} 3 \\ 3 \\ 2 \end{pmatrix} + \lambda \begin{pmatrix} -1 \\ 2 \\ 1 \end{pmatrix} + \mu \begin{pmatrix} 2 \\ 0 \\ 1 \end{pmatrix}$$

where  $\lambda$  and  $\mu$  are scalar parameters.

(a) Show that vector  $2\mathbf{i} + 3\mathbf{j} - 4\mathbf{k}$  is perpendicular to  $\Pi$ .

(2)

(b) Hence find a Cartesian equation of  $\Pi$ .

(2)

The line  $l$  has equation

$$\mathbf{r} = \begin{pmatrix} 4 \\ -5 \\ 2 \end{pmatrix} + t \begin{pmatrix} 1 \\ 6 \\ -3 \end{pmatrix}$$

where  $t$  is a scalar parameter.

The point  $A$  lies on  $l$ .

Given that the shortest distance between  $A$  and  $\Pi$  is  $2\sqrt{29}$

(c) determine the possible coordinates of  $A$ .

(4)

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5. The line  $l_1$  has equation  $\frac{x+5}{1} = \frac{y+4}{-3} = \frac{z-3}{5}$

The plane  $\Pi_1$  has equation  $2x + 3y - 2z = 6$

(a) Find the point of intersection of  $l_1$  and  $\Pi_1$  (2)

The line  $l_2$  is the reflection of the line  $l_1$  in the plane  $\Pi_1$

(b) Show that a vector equation for the line  $l_2$  is

$$\mathbf{r} = \begin{pmatrix} -7 \\ 2 \\ -7 \end{pmatrix} + \mu \begin{pmatrix} 10 \\ 6 \\ 2 \end{pmatrix}$$

where  $\mu$  is a scalar parameter. (5)

The plane  $\Pi_2$  contains the line  $l_1$  and the line  $l_2$

(c) Determine a vector equation for the line of intersection of  $\Pi_1$  and  $\Pi_2$  (2)

The plane  $\Pi_3$  has equation  $\mathbf{r} \cdot \begin{pmatrix} 1 \\ 1 \\ a \end{pmatrix} = b$  where  $a$  and  $b$  are constants.

Given that the planes  $\Pi_1$ ,  $\Pi_2$  and  $\Pi_3$  form a sheaf,

(d) determine the value of  $a$  and the value of  $b$ . (3)

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8.

$$\mathbf{A} = \begin{pmatrix} 3 & 1 & -1 \\ 1 & 1 & 1 \\ k & 3 & 6 \end{pmatrix} \quad k \neq 0$$

- (a) Find, in terms of  $k$ ,  $\mathbf{A}^{-1}$  (4)
- (b) Determine, in simplest form in terms of  $k$ , the coordinates of the point where the following planes intersect.

$$3x + y - z = 3$$

$$x + y + z = 1$$

$$kx + 3y + 6z = 6$$
(3)



7.

$$\mathbf{M} = \begin{pmatrix} 2 & -1 & 1 \\ 3 & k & 4 \\ 3 & 2 & -1 \end{pmatrix} \quad \text{where } k \text{ is a constant}$$

(a) Find the values of  $k$  for which the matrix  $\mathbf{M}$  has an inverse.

(2)

(b) Find, in terms of  $p$ , the coordinates of the point where the following planes intersect

$$2x - y + z = p$$

$$3x - 6y + 4z = 1$$

$$3x + 2y - z = 0$$

(5)

(c) (i) Find the value of  $q$  for which the set of simultaneous equations

$$2x - y + z = 1$$

$$3x - 5y + 4z = q$$

$$3x + 2y - z = 0$$

can be solved.

(ii) For this value of  $q$ , interpret the solution of the set of simultaneous equations geometrically.

(4)

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6.

$$\mathbf{M} = \begin{pmatrix} k & 5 & 7 \\ 1 & 1 & 1 \\ 2 & 1 & -1 \end{pmatrix} \quad \text{where } k \text{ is a constant}$$

(a) Given that  $k \neq 4$ , find, in terms of  $k$ , the inverse of the matrix  $\mathbf{M}$ .

(4)

(b) Find, in terms of  $p$ , the coordinates of the point where the following planes intersect.

$$2x + 5y + 7z = 1$$

$$x + y + z = p$$

$$2x + y - z = 2$$

(3)

(c) (i) Find the value of  $q$  for which the following planes intersect in a straight line.

$$4x + 5y + 7z = 1$$

$$x + y + z = q$$

$$2x + y - z = 2$$

(ii) For this value of  $q$ , determine a vector equation for the line of intersection.

(7)

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8. Two birds are flying towards their nest, which is in a tree.

Relative to a fixed origin, the flight path of each bird is modelled by a straight line.

In the model, the equation for the flight path of the first bird is

$$\mathbf{r}_1 = \begin{pmatrix} -1 \\ 5 \\ 2 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ a \\ 0 \end{pmatrix}$$

and the equation for the flight path of the second bird is

$$\mathbf{r}_2 = \begin{pmatrix} 4 \\ -1 \\ 3 \end{pmatrix} + \mu \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix}$$

where  $\lambda$  and  $\mu$  are scalar parameters and  $a$  is a constant.

In the model, the angle between the birds' flight paths is  $120^\circ$

- (a) Determine the value of  $a$ .

(4)

- (b) Verify that, according to the model, there is a common point on the flight paths of the two birds and find the coordinates of this common point.

(5)

The position of the nest is modelled as being at this common point.

The tree containing the nest is in a park.

The ground level of the park is modelled by the plane with equation

$$2x - 3y + z = 2$$

- (c) Hence determine the shortest distance from the nest to the ground level of the park.

(3)

- (d) By considering the model, comment on whether your answer to part (c) is reliable, giving a reason for your answer.

(1)

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